Inheritance - Is-A-Relationship

Inheritance allows a child class to acquire the attributes & methods of a parent class - just like a child inherits properties from parents in real life.

Parent class	child class	Benifit
- Base class - Super class	- Derived class - sub class	- Code Reusability
The one that gives	The one that receives	- DRY Don't repeat yourself

(Violates DRY)

without inheritance

```
class Student:
    def login(self):          #R
    def register(self):       #R
    def enroll(self):
    def feedback(self):
```

(follow DRY)

with inheritance

```
class AppUser:
    def login(self):
    def register(self):

class Student(AppUser):
    def enroll(self):
```

```
class Teacher:
    def login(self):
    def register(self):
    def create_course(self
    def ave_queries(
```

```
    def feedback(self):

class Teacher(AppUser):
    def create_course(self):
    def ave_queries(self):
```

```
class AppUser:  # parent

    def __init__(self, username):
        self.username = username

    def login(self):
        print(self.username, "logged in!")

class Student(AppUser):  # child
    def __init__(self, roll_no):
        self.roll_num = roll_no

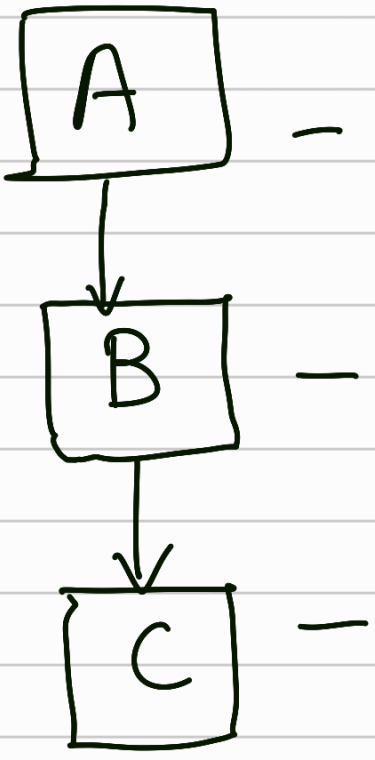
    def enroll(self):
        print("Enrolled Successfully!")

Student1 = Student(100)
Student1.enroll()
Student1.login()
```

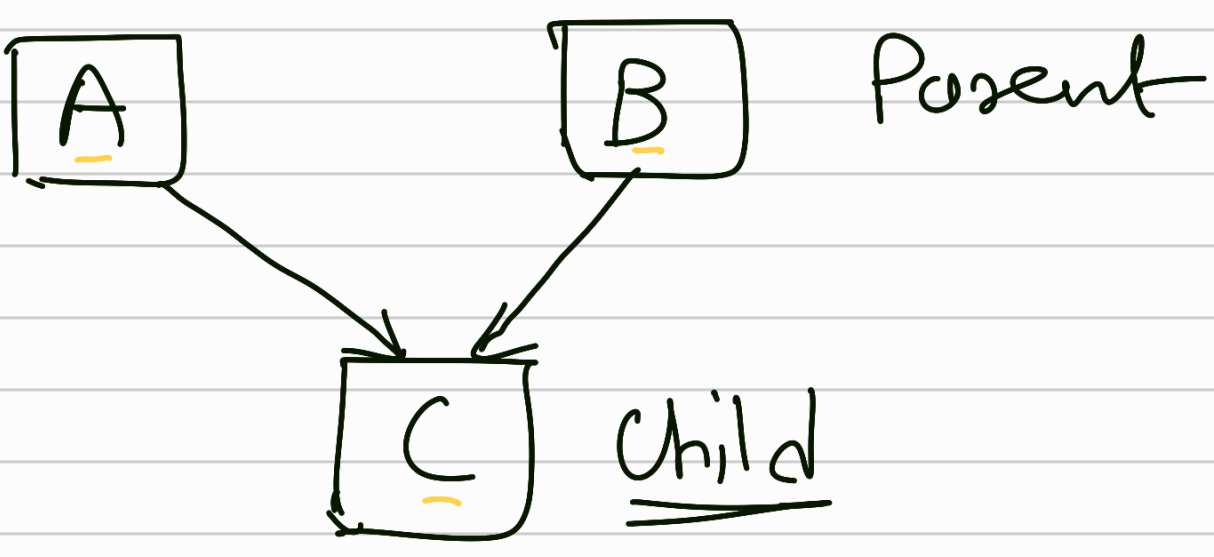
Student1 100

Types of Inheritance in Python.

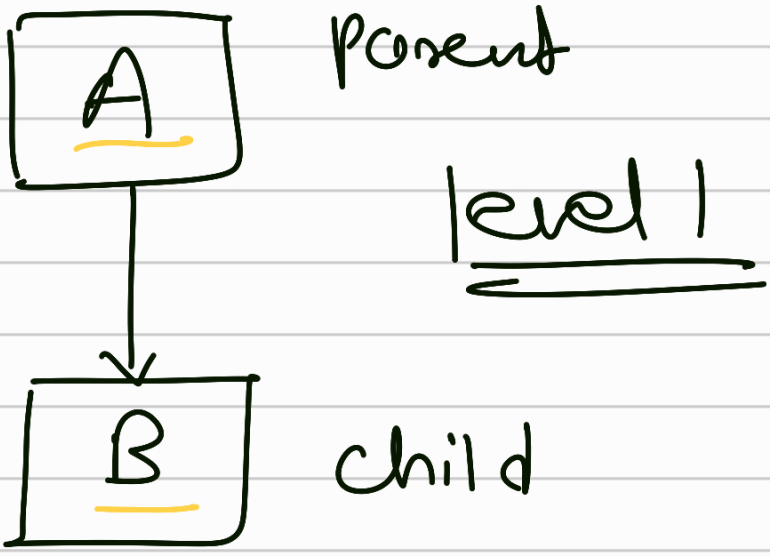
① Multilevel inheritance



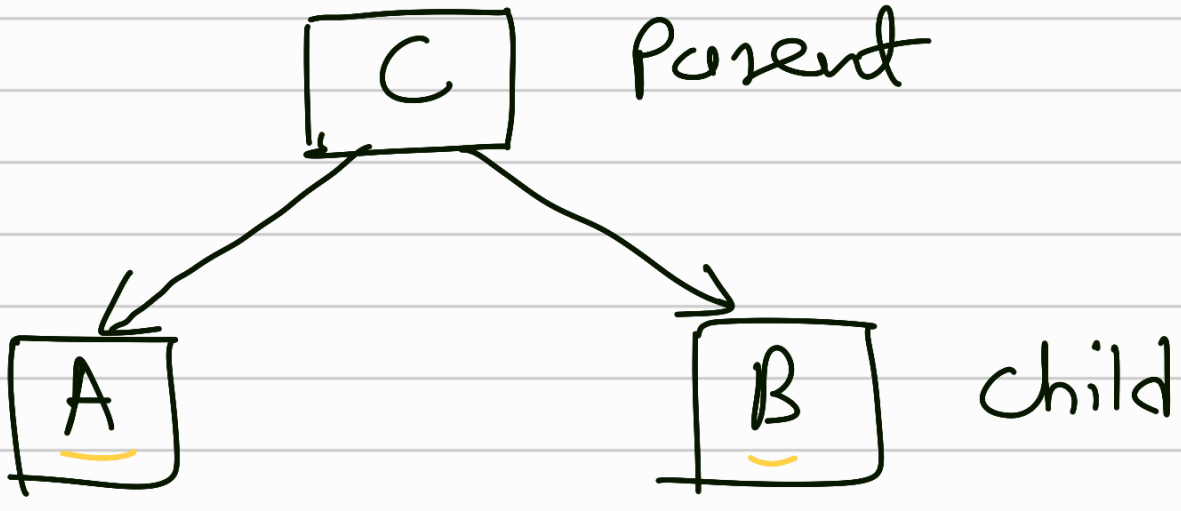
② multiple inheritance



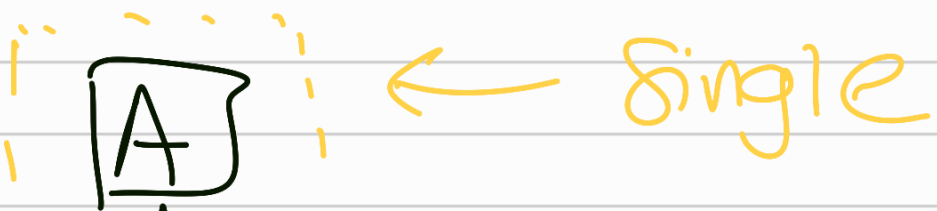
③ Single inheritance

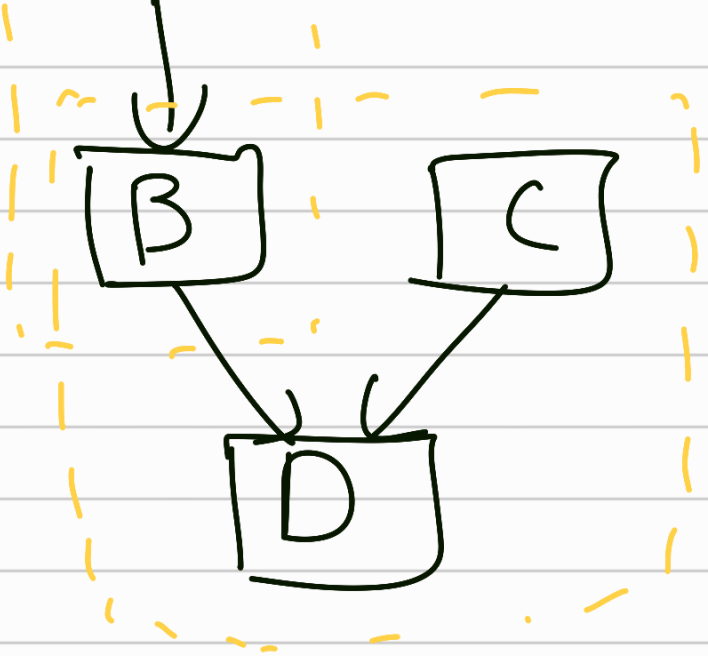


④ Hierarchical inheritance



⑤ Hybrid inheritance





← multiple